Package 'RCamelsPE'

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Type Package

Title Access, Process, and Visualize CAMELS-PE Data

Version 0.1.1

Description Provides tools to read, process, and visualize hydrological, climatic, physiographic, and geospatial data from the CAMELS-PE dataset (Catchment Attributes and Meteorology for Large-sample Studies - Peru). The package is designed to work with externally stored CAMELS-PE data and does not include the dataset itself. Functions support reading daily time series, catchment attributes, station metadata, metadata dictionaries, and geospatial layers, as well as basic visualization of time series and catchment boundaries.

License GPL (>= 2)

Encoding UTF-8

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Imports dplyr,
ggplot2,
readr,
rlang,
sf,
stats,
tools,
utils

Suggests knitr,
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testthat (>= 3.0.0)

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URL `https://github.com/hllauca/RCamelsPE`

BugReports `https://github.com/hllauca/RCamelsPE/issues`

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VignetteBuilder knitr

Contents

`get_camels_path` *Get CAMELS-PE dataset path*

Description

Get CAMELS-PE dataset path

Usage

```
get_camels_path()
```

Value

Character path to the CAMELS-PE root directory.

`plot_attribute_map` *Plot CAMELS-PE Attribute Map*

Description

Creates a thematic map of CAMELS-PE catchments using a selected attribute. The function joins attribute data with catchment geometries by `gauge_id` and returns a `ggplot` object.

Usage

```
plot_attribute_map(
  catchments,
  attributes,
  variable,
  gauges = NULL,
  na_color = "grey80",
  ...
)
```

Arguments

`catchments`	An `sf` object with catchment polygons.
`attributes`	A data frame or tibble with CAMELS-PE attributes.
`variable`	Character string. Name of the attribute to visualize.
`gauges`	Optional `sf` object with gauge point locations.
`na_color`	Character string. Color for missing values.
`...`	Additional arguments passed to `ggplot2::geom_sf()` for catchments.

Value

A `ggplot` object.

Examples

```
## Not run:
catchments <- read_geospatial("catchments")
gauges <- read_geospatial("gauges")
attr <- read_attributes("topographic")

plot_attribute_map(
  catchments = catchments,
  attributes = attr,
  variable = "area_km2",
  gauges = gauges
)

## End(Not run)
```

`plot_catchments` *Plot CAMELS-PE Catchments*

Description

Creates a map of CAMELS-PE catchments and optionally overlays gauge locations. If `gauge_id` is provided, only one catchment and its outlet are plotted, with the gauge ID shown as a panel header.

Usage

```
plot_catchments(catchments, gauges = NULL, gauge_id = NULL, fill = NULL, ...)
```

Arguments

`catchments`	An `sf` object with catchment polygons.
`gauges`	Optional `sf` object with gauge point locations.
`gauge_id`	Optional character string. Gauge ID used to filter one catchment and its gauge before plotting. If `NULL`, all catchments are plotted.
`fill`	Optional character string. Name of a catchment column used to fill polygons. If `NULL`, a constant fill is used.
`...`	Additional arguments passed to `ggplot2::geom_sf()`.

Value

A ggplot object.

Examples

```
## Not run:
catchments <- read_geospatial("catchments")
gauges <- read_geospatial("gauges")

# Plot all catchments
plot_catchments(catchments, gauges)

# Plot one catchment and its outlet
plot_catchments(catchments, gauges, gauge_id = "PE_221804")

## End(Not run)
```

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plot_timeseries *Plot CAMELS-PE Time Series*

Description

Creates a time series plot for one CAMELS-PE variable, such as precipitation, observed streamflow, simulated streamflow, or temperature. The function returns a `ggplot` object, so it can be further customized using standard `ggplot2` layers.

Usage

```
plot_timeseries(
  data,
  variable = "flow_obs",
  gauge_id = NULL,
  date_col = "date",
  facet = TRUE,
  scales = "free_y",
  ...
)
```

Arguments

`data`	A data frame or tibble containing CAMELS-PE time series.
`variable`	Character string. Name of the variable to plot.
`gauge_id`	Optional character vector. Gauge IDs used to filter the data before plotting. If `NULL`, all available gauges are plotted.
`date_col`	Character string. Name of the date column.
`facet`	Logical. If `TRUE`, creates one panel per gauge ID.
`scales`	Character string. Scales passed to `ggplot2::facet_wrap()`. One of `"fixed"`, `"free"`, `"free_x"`, or `"free_y"`.
`...`	Additional arguments passed to `ggplot2::geom_line()`.

Value

A ggplot object.

Examples

```
## Not run:
ts <- read_timeseries(
  gauge_id = c("PE_250101", "PE_200907"),
  vars = c("date", "gauge_id", "prec", "flow_obs", "flow_sim", "tmean")
)

plot_timeseries(ts, variable = "flow_obs")
plot_timeseries(ts, variable = "prec", linewidth = 0.4)
plot_timeseries(ts, variable = "flow_obs", facet = FALSE)
plot_timeseries(ts, variable = "flow_obs", gauge_id = "PE_250101")

## End(Not run)
```

---

read_attributes *Read CAMELS-PE Catchment Attributes*

Description

Reads catchment attributes from the CAMELS-PE dataset. Attributes are stored in thematic groups, including topography, climate, hydrological signatures, land cover, geology, soil, and human intervention.

Usage

```
read_attributes(type = "all", gauge_id = NULL, path = get_camels_path())
```

Arguments

type	Character string. Attribute group to read. One of "topographic", "climatic", "hydrological", "landcover", "geologic", "soil", "human_intervention", or "all". Default is "all".
gauge_id	Character vector or NULL. Optional gauge identifiers used to filter the returned attributes, for example "PE_0001" or c("PE_0001", "PE_0002"). If NULL, all available catchments are returned.
path	Character string. Optional path to the CAMELS-PE root directory. If not provided, the path previously defined with set_camels_path() is used.

Details

The function can read one attribute group at a time or join all available groups by gauge_id. To inspect the available attribute names, descriptions, units, and thematic groups, use read_attribute_dictionary().

Available attribute groups are:

- "topographic": topographic and physiographic attributes.
- "climatic": climatic indices and long-term climate descriptors.

- "hydrological": hydrological signatures.
- "landcover": land-cover attributes.
- "geologic": geological attributes.
- "soil": soil attributes.
- "human_intervention": human intervention and regulation attributes.
- "all": all available attribute groups joined by gauge_id.

Attribute files are expected to be stored in 02_attributes/. Each attribute file must contain a gauge_id column. When type = "all", missing attribute files are skipped with a warning and all available files are joined. If no valid attribute files are found, the function stops with an error.

Expected files are:

- topographic_attributes.csv
- climatic_indices.csv
- hydrological_signatures.csv
- landcover_attributes.csv
- geologic_attributes.csv
- soil_attributes.csv
- human_intervention_attributes.csv

Value

A tibble with CAMELS-PE catchment attributes. The output includes one row per catchment and a gauge_id column used as the main identifier. When type = "all", attribute groups are joined by gauge_id.

Examples

```
## Not run:
set_camels_path("D:/DATA/CAMELS-PE")

# Inspect available attributes
attr_dict <- read_attribute_dictionary()
head(attr_dict)

# Read topographic attributes
topo <- read_attributes(type = "topographic")

# Read climatic attributes for selected catchments
clim <- read_attributes(
  type = "climatic",
  gauge_id = c("PE_250101", "PE_200907")
)

# Read all available attribute groups
attr_all <- read_attributes(type = "all")

## End(Not run)
```

read_geospatial *Read CAMELS-PE Geospatial Data*

Description

Reads geospatial data from the CAMELS-PE dataset, including gauge locations and catchment boundaries. The data are stored as GeoPackage files in the `04_geospatial` directory.

Usage

```
read_geospatial(type = c("gauges", "catchments"), path = get_camels_path())
```

Arguments

`type`	Character string. Either `"gauges"` or `"catchments"`.
`path`	Character string. Optional path to the CAMELS-PE root directory. If not provided, the path previously defined with `set_camels_path()` is used.

Details

Available types:

- `"gauges"`: point locations of gauging stations.
- `"catchments"`: polygon boundaries of catchments.

Value

An `sf` object.

Examples

```
## Not run:
set_camels_path("D:/DATA/CAMELS-PE")

# Read gauges
gauges <- read_geospatial("gauges")

# Read catchments
catchments <- read_geospatial("catchments")

# Plot
plot(catchments$geometry)
plot(gauges$geometry, add = TRUE, col = "red")

## End(Not run)
```

`read_metadata` *Read CAMELS-PE Station Metadata*

Description

Reads the station metadata table from the CAMELS-PE dataset. This file contains the main information for each gauging station, such as the station identifier, name, location, drainage area, river, basin, or other available metadata fields depending on the CAMELS-PE dataset version.

Usage

```
read_metadata(path = get_camels_path())
```

Arguments

`path`	Character string. Optional path to the CAMELS-PE root directory. If not provided, the path previously defined with `set_camels_path()` is used.

Details

The function reads the file:

`01_metadata/stations.csv`

from the CAMELS-PE root directory.

Value

A tibble with CAMELS-PE station metadata.

Examples

```
## Not run:
# Define the CAMELS-PE dataset path
set_camels_path("D:/DATA/CAMELS-PE")

# Read station metadata
stations <- read_metadata()

# Preview the first rows
head(stations)

## End(Not run)
```

`read_timeseries` *Read CAMELS-PE Time Series*

Description

Reads hydrometeorological time series from the CAMELS-PE dataset. The function supports two reading modes:

Usage

```
read_timeseries(
  gauge_id = NULL,
  global = FALSE,
  vars = NULL,
  path = get_camels_path()
)
```

Arguments

`gauge_id`	Character vector or `NULL`. Gauge identifiers to read, for example `"PE_0001"` or `c("PE_0001", "PE_0002")`. This argument is required when `global = FALSE`. When `global = TRUE`, it can be used to filter the global time series file.
`global`	Logical. If `TRUE`, reads the global file `03_timeseries/timeseries.csv`. If `FALSE`, reads individual catchment files from `03_timeseries/by_catchment/`. Default is `FALSE`.
`vars`	Character vector or `NULL`. Optional variable names to keep in the returned table. If `NULL`, all available variables are returned.
`path`	Character string. Optional path to the CAMELS-PE root directory. If not provided, the path set by `set_camels_path()` is used.

Details

- **Global mode**: reads the full time series table from `03_timeseries/timeseries.csv`.
- **By-catchment mode**: reads one or more catchment-specific files from `03_timeseries/by_catchment/`. This is the recommended mode for large datasets because only the requested catchments are loaded.

The dataset must be available in a local directory previously defined with `set_camels_path()`, or supplied directly through the `path` argument.

Available variables typically include:

- `"date"`: date of observation.
- `"gauge_id"`: catchment identifier.
- `"prec"`: precipitation (mm/day).
- `"prec_var"`: precipitation variability or variance (mm^2/day^2).
- `"flow_obs"`: observed streamflow (mm/day).
- `"flow_sim"`: simulated streamflow (mm/day).
- `"tmean"`: mean air temperature (°C).

- "tmax": maximum air temperature (°C).
- "tmin": minimum air temperature (°C).
- "pet": potential evapotranspiration (mm/day).
- "srad": solar radiation (MJ/m^{2}day).
- "vprp": vapor pressure (hPa).

Available variables may differ depending on the CAMELS-PE dataset version.

In by-catchment mode, each requested gauge is expected to have a file named <gauge_id>.csv inside 03_timeseries/by_catchment/. For example, PE_0001 is expected to be stored as 03_timeseries/by_catchmen

If one or more requested files are missing, the function issues a warning and reads the files that are available. If no valid files are found, the function stops with an error.

Value

A tibble containing CAMELS-PE time series data. The output usually includes at least date, gauge_id, and the selected hydrometeorological variables. If a date column is available, it is converted to class Date.

Examples

```
## Not run:
set_camels_path("D:/DATA/CAMELS-PE")

# Read all variables for one catchment
ts1 <- read_timeseries(gauge_id = "PE_221804")

# Read selected variables for multiple catchments
ts2 <- read_timeseries(
  gauge_id = c("PE_250101", "PE_200907"),
  vars = c("date", "gauge_id", "prec", "flow_obs")
)

# Read the global time series file
ts_all <- read_timeseries(global = TRUE)

# Read the global file and filter selected gauges
ts_sel <- read_timeseries(
  gauge_id = c("PE_250101", "PE_200907"),
  global = TRUE
)

## End(Not run)
```

---

set_camels_path *Set CAMELS-PE dataset path*

Description

Set CAMELS-PE dataset path

Usage

```
set_camels_path(path)
```

Arguments

`path`	Character. Path to the CAMELS-PE root directory.

Value

Invisibly returns the normalized path.

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